The embryos at 8-32 celled stages are recovered and transferred to a surrogate mother. For another round of ovulation, the same genetic mother is utilized. This technology can be applied to cattle, sheep and buffaloes. Advantage of this technology is to produce high milk yielding females and high-quality meat yielding bulls in a short time.

Breeds of Dairy animals

Dairying is the production and marketing of milk and its products. Dairy operation consists of proper maintenance of cattle, the collection and processing the milk and its by products. There are 26 well defined breeds of cattle and 6 breeds of buffaloes in India. Cattles are classified under three groups based on the purpose they serve to man (Figure 12. 12). They are

i. **Dairy breeds or Milch breeds**: They are high milk yielders with extended lactation. Eg., Sindhi, Gir, Sahiwal, Jersy, Brown Swiss, Holstein cattle.

ii. **Draught purpose breeds**: Bullocks are good for draught purpose. Eg. Kangayam, Malvi

iii. **Dual Purpose breeds**: Cows are meant for yielding more milk and bullocks are used for better drought purpose. Eg. Ongole, Hariana



Vechur breed is the smallest breed of Cow as per World Guinness Records.

Average length: 124cms

Average height: 87 cms

Origin: Vechur village, Kottayam District of Kerala

It produces large amount of milk in relation to the food consumption

India is the largest producer of Milk, globally.

India has many popular breeds of cows and buffaloes.

Prominent indigenous cow breeds in India - Gir, Red sindhi, Sahiwal, Hallikar, Amritmahal, Khillari, Kangayam, Bargur, Umblachery, Pulikulam, Alambadi, Tharparkar, Haryana, Kankrej, Ongole, Krishna valley and Deoni.

To meet the milk demand of the growing population, milk breeds are preferred by farmers in small scale farms. Goats are also used all over India for supplementing deficiencies in milk production. Some of the breeds of cattle that are good milkers are Jamunapari in Ganga-Jamuna riverine tracts, Beetal in Punjab, Bar–bari in Uttarpradesh.



Gir Kangeyam Ongole

Figure 12.12 Different breeds of cattle

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